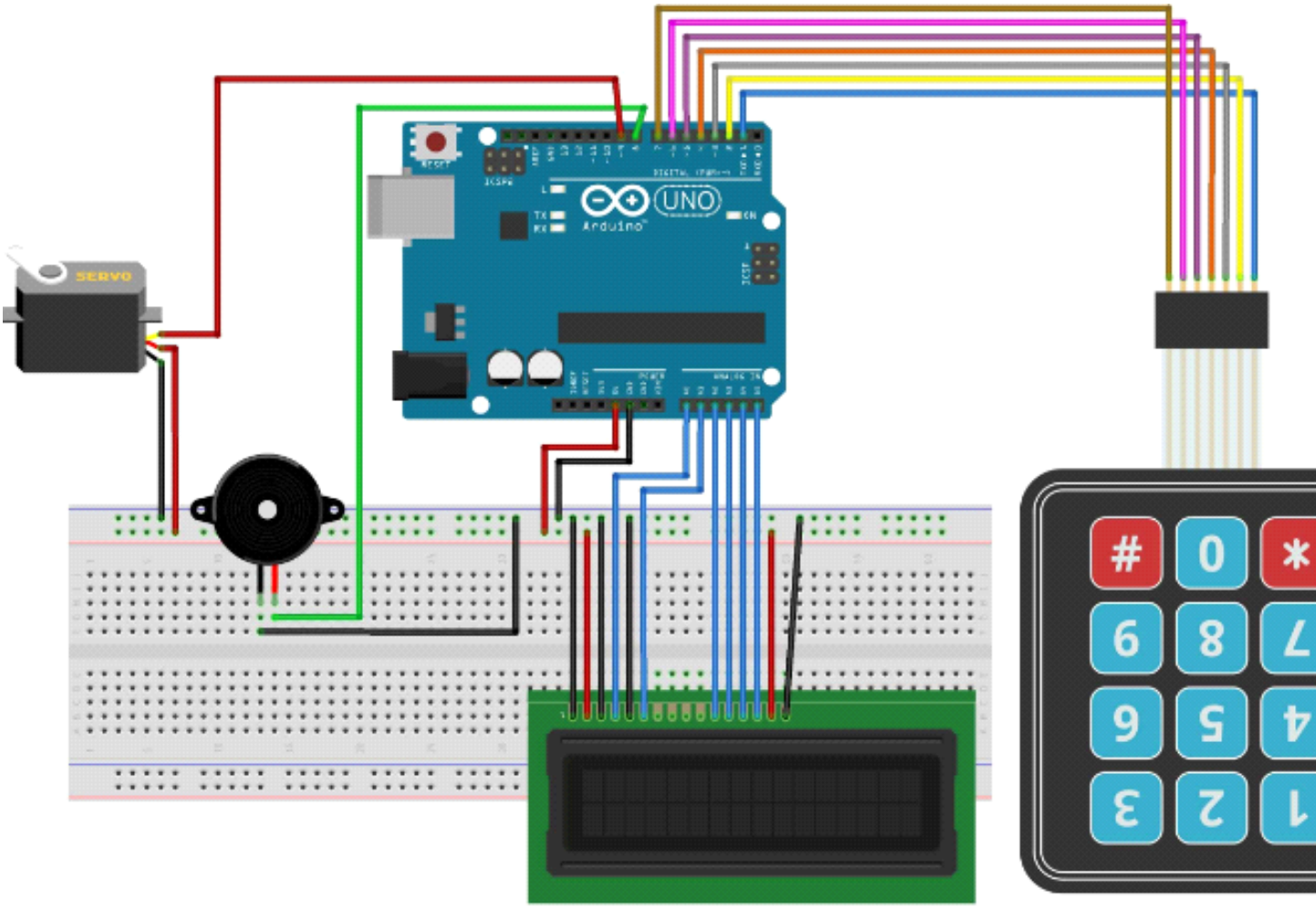
SriLankan Institute of Information technology

IT 1140-Fundementals of Computing

Year 1, Semester 1,2024

Name	IT number
Frank	IT24104354
Thirath	IT24104295
Sathsare	IT24104328
Sanduni	IT24201283
Randi	IT24104341
Abhimani	IT24103927

Circuit Diagram



- Work Breakdown Structure

1. IT24104354 (Frank) : connected the arduino and the other components and gave instructions to code
2. IT24104295 (Thirath) : Focused on the coding errors and fixed it
3. IT24201283 (sanduni) : Helped to build the hardware part of the system

4. IT24103927 (abhimani) : Briefly studied about the project output
5. IT24104341 (Randi) : identfied about the components of the system
6. IT24104328 (sathsara) : gave ideas about the arduino software part

Arduino Based Door Lock and Light control System

Lab Report

- **Introduction**

The Arduino-Based Entryway Lock and Light Control Framework is a microcontroller-driven project intended to upgrade home robotization, security, and energy productivity. The framework coordinates an entryway locking system and computerized light control utilizing Arduino, sensors, and actuators. It exhibits the utilization of implanted frameworks in certifiable applications to work on everyday schedules and further develop security

- **Objectives**
- To foster a protected entryway locking system that can be controlled electronically.
- To carry out a robotized light control framework that answers natural or client input.
- To investigate and show the utilization of Arduino microcontrollers in home computerization.

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- **Usable Components**

  - Arduino Uno
  - LCD Display (16*2) with I2C module
  - Keypad
  - Servo Motor
  - Jumper Wires
  - Buzzer(5v)
  - PIR Sensor
  - LDR Sensor
  - LED Bulbs

- **Methodology**

  - System Design

- The entryway lock framework utilizes a keypad module to include a password, confirmed by Arduino. On the off chance that the code is right, the servo engine opens the entryway; in any case, the ringer enacts as an advance notice.

- The light control framework utilizes an LDR to identify surrounding light levels. On the off chance that the light level falls under a limit, the transfer enacts the Drove bulb.

- **Workflow**

- **Entryway Lock Framework**

1. The client inputs a password through the keypad.

2. Arduino contrasts the entered code and the pre-modified code.

3. On the right information, the servo engine pivots to open the entryway.

4. On wrong information, the signal sounds to demonstrate a blunder.

- **Light Control Framework**

1. The LDR consistently screens the surrounding light power.

2. On the off chance that the light power dips under a predefined esteem, the Arduino triggers the hand-off module to turn on the Drove bulb.

3. Assuming the surrounding light increments over the limit, the light switches off naturally.

- **Challanges Faced**

1. **Power Supply**
2. **Code Debugging**
3. **Hardware Integration**

- **Results**

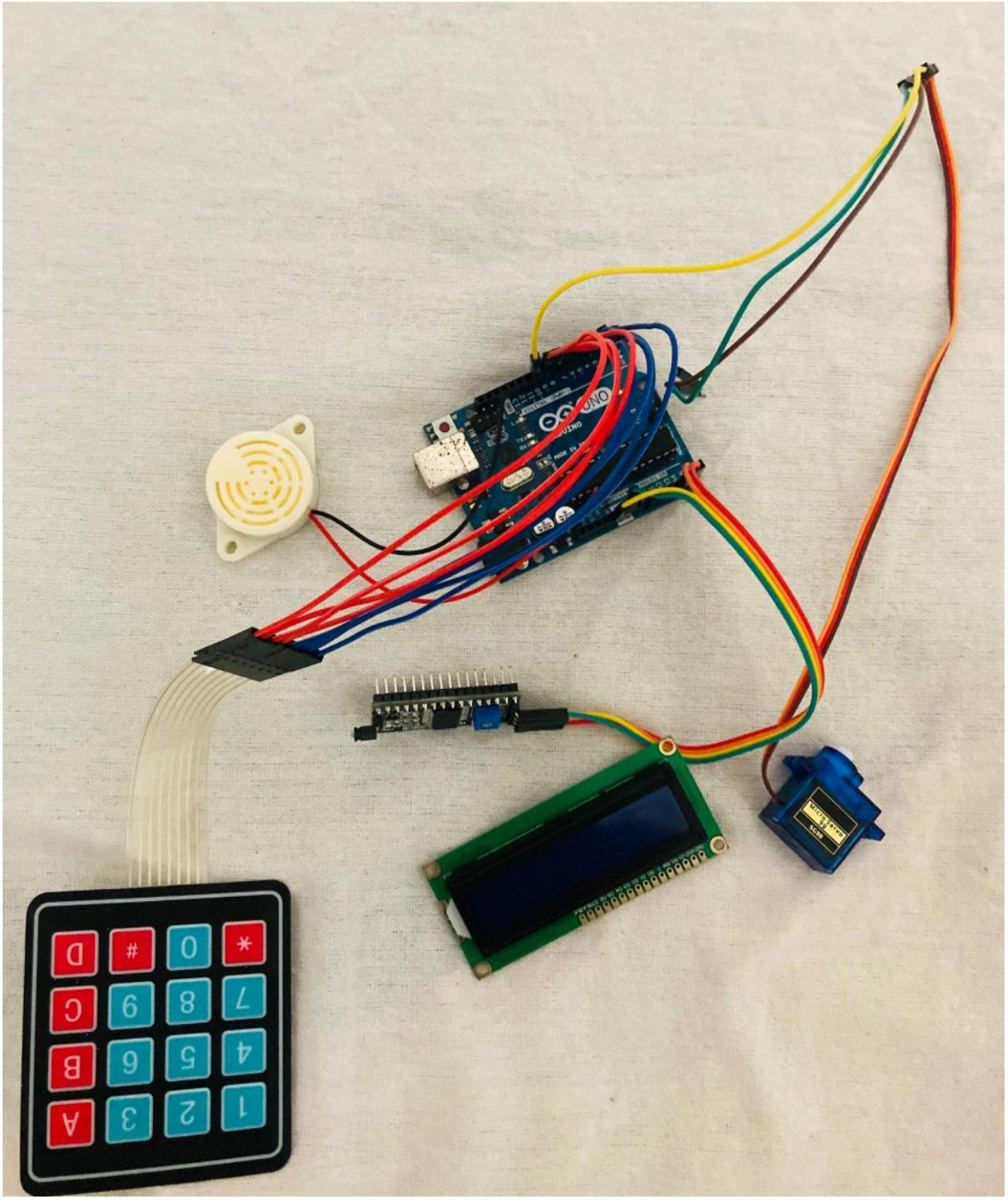
1. **The Door Lock System Successfully Allowed Access Upon entering the correct password**

2. **Light control System Efficiently Switched the Led Bulb on and Off Baced Components**

- **References**

1. **Arduino Official Documentation**
2. **Tutorial Point**
3. **ChatGPT, YouTube, Google**

Proof Of Work



Conclusion

The Arduino-Based Entryway Lock and Light Control Framework is a successful and reasonable execution of home robotization. The venture features the capability of microcontrollers in rearranging and getting everyday assignments. Future upgrades could incorporate coordinating remote control by means of a portable application or adding biometric verification for improved security.